

SEID 2364 – Assignment 2 Defining Intelligence

Part 1 – Initial Speculative Definition of Intelligence

Initial Provocations from Class

1. What makes a system intelligent? 2. Can machines truly think? 3. Is intelligence only connected to humans? 4. Does intelligence require consciousness?

Week 2: Initial Speculative Definition of Intelligence

My initial definition of intelligence is the ability of a system to learn, adapt, solve problems, and respond to changing situations in ways that help achieve goals. Intelligence is not limited to humans because machines can also process information, make predictions, and improve performance through data and experience. However, human intelligence still includes emotions, creativity, and social understanding in ways machines may not fully possess.

Part 2 – Core Readings Reflection

After reviewing the readings from Legg and Hutter, the Stanford Encyclopedia of Philosophy, and Alan Turing's "Computing Machinery and Intelligence," I noticed that intelligence is defined differently depending on the field of study. Some definitions focus on reasoning and problem-solving, while others focus on adaptation, learning, and goal achievement.

Legg and Hutter attempted to combine many definitions into the idea of "Universal Intelligence," which describes intelligence as the ability of an agent to achieve goals across many environments. I found this useful because it applies to both humans and machines.

The Stanford Encyclopedia emphasized philosophical questions about cognition, reasoning, and consciousness. This made me realize that intelligence is not only about producing correct answers but also about understanding and interpretation.

Turing approached intelligence from a practical perspective. Instead of asking whether machines can "think," he asked whether machine behavior could appear intelligent to humans. I think this changed the discussion about AI by focusing on observable behavior rather than internal consciousness.

Part 3 – Rule Set for Intelligence

Based on the readings, I think a system can be considered intelligent if it:

1. Learns from information or experience 2. Adapts to different environments 3. Solves problems or achieves goals 4. Responds to new situations 5. Uses information to guide decisions

Boundary Testing

A calculator can solve mathematical problems very accurately, but it does not adapt or learn, so I would not fully classify it as intelligent.

An AI chatbot can learn patterns, generate responses, and adapt to conversations, so it fits more closely with my definition of intelligence.

Humans remain different because human intelligence also includes emotional awareness, moral reasoning, and creativity beyond purely computational tasks.

Conclusion

This assignment showed me that intelligence is not a single fixed concept. Different definitions focus on different abilities such as learning, reasoning, adaptation, or behavior. I now think intelligence should be understood as a flexible model rather than a strict category, especially when comparing human and machine systems.